Model Development Phase Template

Date	25 February 2026
Team ID	259422
Project Title	Online Payments Fraud Detection using Machine Learning
Maximum Marks	6 Marks

Model Selection Report

Due to severe class imbalance (~0.13% fraud cases), model selection was based on:

- Recall (Fraud Detection Rate) – Primary metric
- F1 Score – Balanced performance indicator
- Confusion Matrix analysis
- Generalization capability
- Computational efficiency

Accuracy was considered secondary because it can be misleading in imbalanced datasets.

Model Selection Report:

Model	Description	Hyperparameters	Performance Metric
Decision Tree Classifier	Single tree-based classifier that splits data based on feature thresholds. Acts as baseline model.	`max_depth=10`, `min_samples_split=5`, `class_weight='balanced'`, `random_state=42`	Accuracy: 99.70% Recall (Fraud): 0.89 F1 Score: 0.89
Extra Trees Classifier	Ensemble of randomized decision trees; introduces higher randomness in splits for variance reduction and faster computation.	`n_estimators=100`, `max_depth=None`, `class_weight='balanced'`, `random_state=42`	Accuracy: 99.97% Recall (Fraud): 0.78 F1 Score: 0.87
Random Forest Classifier	Ensemble model using multiple decision trees with bootstrap aggregation to reduce variance and improve robustness.	`n_estimators=100`, `max_depth=None`, `class_weight='balanced'`, `random_state=42`	Accuracy: 99.97% Recall (Fraud): 0.79 F1 Score: 0.87